

Receding Horizon and Optimization-based Control for UAV path planning with Collision Avoidance

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Abstract

This paper proposes an online energy-efficient path planning approach for UAVs in complex environments. The path planning problem is formulated as a minimization optimization problem based on Mixed Integer Linear Programming (MILP), where a cost function is designed to minimize energy consumption while ensuring terrain obstacle avoidance within a limited detection range. To achieve this, we apply a Receding Horizon Control (RHC) and optimization approach. The entire path is divided into segments or sub-paths, with constraints in place to prevent collisions with obstacles. This proposed optimization approach enables fast navigation through dense environments, ensuring a collision-free path. For further optimizing the path for energy, a path smoothing strategy is introduced to reduce energy consumption caused by sharp turns. The results demonstrate the effectiveness and accuracy of the proposed approach in dense environments with a high risk of collisions with obstacles.

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1. Introduction

Due to the rapid advancement of technology, unmanned aerial vehicle (UAV) has gained significant attention. It offers remarkable attributes, such as providing additional degrees of freedom for controlled mobility and enabling instant and flexible deployment [1, 2]. Thus, UAV has been applied in many military and commercial applications, such as traffic control, aerial surveillance, area coverage, search and rescue, disaster management, traffic policing, and precision agriculture [3, 4, 5, 6]. In such applications, the demand for more sophisticated task often requires the sophisticated strategy for the UAV to accomplish its task efficiently.

One of the main essential aspects of UAV is path planning, which is one of the core technologies that plays a key role in completing UAV missions. It involves finding an optimal path from a starting point to a destination with a

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predefined objective [7]. Several advantages have been provided by path planning, including cost and time efficiency, reduced human intervention, resource optimization, the ability to collect and analyse data, scalability, and adaptability. These advantages make it a valuable tool in various applications, ranging from industrial automation and agriculture to search and rescue operations and environmental monitoring.

A mixed integer linear programming (MILP) is applied in several approaches to address multiple constraints [8, 9, 10]. For example, dynamic programming and branch and bound are exact optimization methods applied to handle the problem optimally via developing a formulation of MILP and solving the problem within a short interval in a drone point of view. The MILP formulation was considered for UAV guidance problems for several applications.

A UAV path optimization problem is formulated as a MILP and solved via CPLEX solver in [11], in which the path optimization is addressed. In [12], a UAV path planning problem is formulated as a maximization optimization problem based on MILP where target tracking performance is the goal to be maximized. In [13], A coverage path planning algorithm was introduced to address the coverage path planning problem. The objective was to generate a collision-free path that covers the entire region while minimizing energy consumption. The problem was formulated based on MILP, with the aim of minimizing the energy consumption.

This research paper focuses on the development of a novel approach for guiding a UAV through an obstacle-filled environment. The proposed approach tackles the problem of path planning by utilizing MILP, which determines the waypoints for the UAV. The primary contribution of the paper is the use of CPLEX solver to obtain an exact solution for the MILP within drone field of view. To this end, the concept of a dynamic finite horizon is implemented, which solves the MILP within a local spatial region that continuously updates as the UAV progresses along its path. To minimize energy consumption, a cost function is proposed that incorporates problem constraints. The UAV aims to reach the desired destination while minimizing the energy consumption. The constraint of maintaining a collision avoidance is imposed on the UAV throughout the entire flight duration to ensure the navigation from the initial to the final point while minimizing the cost function. Notably, we apply the energy model derived in [14] from real experiments that considered different maneuvering activities such as flying horizontal, vertical, hovering in place, and taking a turn.

The contributions of this work can be summarized as follow.

1. An energy-efficient UAV path planning approach is introduced with the objective of minimizing energy consumption while avoiding collision with terrain obstacles. The concept of a dynamic finite horizon is implemented, which solves the MILP within a local spatial region that continuously updates the UAV location along its path as it progresses.
2. Constraints of path planning problem with the goal of minimizing overall energy consumption are derived and analyzed, and the MILP based formulation is designed to find the optimal trajectory for UAV from starting location to the destination while satisfying all constraints. These constraints ensure unique waypoints along the path and provide safety measures for returning to recharge in case the drone has insufficient energy to reach the destination.
3. A smoother strategy is applied on the generated path to smooth the turns and reduce the energy consumption during taking turns.

The paper is structured as follows. In Section 2, we present our proposed energy-efficient UAV path planning approach. This section covers collision avoidance, MILP based formulation of the problem, discussion, and analysis of the optimization problem, and path smoother strategy. Section 3 presents the results and corresponding discussions. Finally, the work is concluded in Section 4.

2. Energy-efficient Path Planning Approach

The concept behind the RHC method is to create a moving time window at each sampling interval, obtain the optimal solution within that window, and then shift the window forward based on a predefined time step. RHC's real-time capabilities and optimization give it significant advantages in an online path planning. Moreover, a finite horizon is employed to restrict the area around the UAV. It is designed to be sufficiently large to allow the UAV to maneuver

and avoid collisions while keeping the computational time manageable. The duration required for collision avoidance depends on the UAV's velocity and the predicted collision position. Fig. 1 shows a schematic diagram of the UAV navigation path through environment filled by obstacles. The horizon is defined by UAV's point of view (defined as a square area) as in the figure.

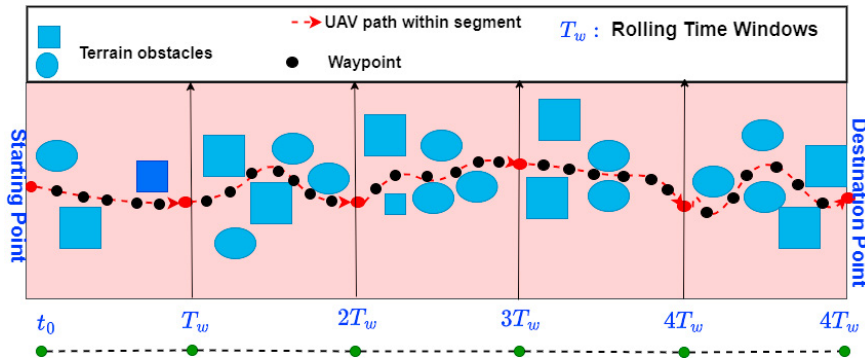


Fig. 1: Path planning diagram of RHC.

The steps of the proposed approach can be summarized are as follows.

- Step 1: At sampling time t_0 , initialize UAV and environment information, construct rolling time window T_w . The environment information involves size of segment to be planned for path and obstacles in the segment.
- Step 2: Decompose the segment area represented by UAV point of view into cells. Then formulate the path planning within the segment as a minimization optimization problem based on MILP and apply optimization objective function for energy consumption minimization.
- Step 3: The problem is then solved by the optimization solver (CPLEX solver), and the optimal path that minimizes the energy consumption while meeting all constraints is generated.
- Step 4: Sooth the generated path and Update the location of a UAV and environment information. In the next sampling, repeat steps from step2 to step4 until the destination is reached.

The block diagram of the proposed approach is depicted in Fig. 2. In the following subsection, we will explain the parts of the proposed approach in detail.

2.1. Collision Avoidance module

To avoid obstacle, we consider involving a safety margin around the obstacle as in [15, 16]. The obstacles constraints can be represented as follows; the path is penalized by high value if it goes through obstacles. This module is responsible for formulating the constraints of obstacles collision in the environment.

2.2. MILP Solution module

This module receives objective and constraints and formulates the problem through the following parts.

2.2.1. Segment Area Decomposition

The given segment area is decomposed to generate all possible waypoints using region decomposition technique. The waypoints that the UAV needs to visit are added to the MILP formulation in the form of constraints. Then, path planning problem is formulated as a minimization problem in which the energy consumption is minimized as much as possible. It has to be stressed that the end point within a segment is the closest point to the actual destination. Finally, the problem is solved using CPLEX solver and the collision-path is generated. Because of the lack of space, we refer the reader to [13, 15] for more information about segmentation process.

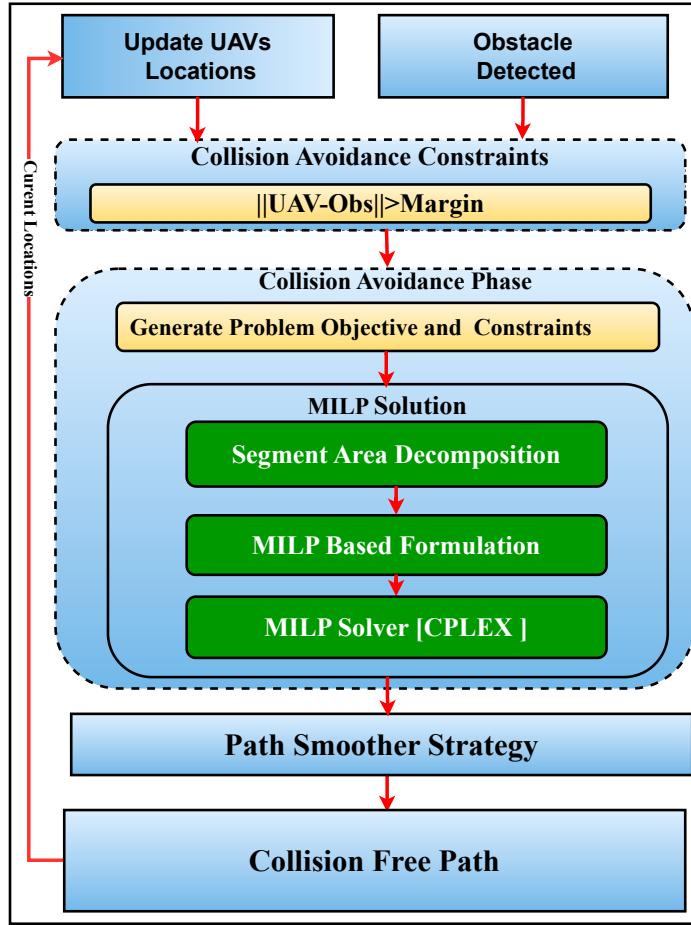


Fig. 2: Block diagram of energy-efficient path planning approach.

2.2.2. MILP Based Formulation

The objective is to navigate through a dense environment, delivering the drone to its destination safely and with high energy efficiency. The decision variables of the target path planning problem are inspired by the classical MILP formulation.

Let N_p is a number of possible locations in the segment defined by drone's point of view, E_u is the available energy of a UAV, E_{ij} is the consumed energy for flying from $position_i$ to $position_j$, and OC_{ij} is the cost of collision with obstacles. The OC_{ij} can be written as follows:

$$OC_{ij} = \begin{cases} \inf & \text{if the path from } position_i \text{ and } position_j \text{ collides with obstacle,} \\ 0 & \text{Otherwise.} \end{cases}$$

The decision variable x_{ij} can be written as follow:

$$x_{ij} = \begin{cases} 1 & \text{if a UAV moves from } position_i \text{ to } position_j, \\ 0 & \text{Otherwise.} \end{cases}$$

For intermediate decision variable, y_i is an additional variable indicating whether position i is visited or not, and can be written as follow

$$y_i = \begin{cases} 1 & \text{if } location_i \text{ is visited,} \\ 0 & \text{Otherwise.} \end{cases}$$

The constraints' aim is to provide a collision-free path to deliver the drone to the destination safely. The objective function considers both path planning and collision avoidance.

In light of this definition, we formulate the optimization problem for minimizing the flight energy consumption by optimizing the path of a UAV, while considering given constraints. Mathematically, it can be expressed as

$$\begin{aligned}
 \min_{\text{Energy}} \quad & \text{ObjFun} = \sum_{i=1}^{N_p} \sum_{j=2, i \neq j}^{N_p} (E_{ij}x_{ij} + OC_{ij}x_{ij}) \\
 \text{s.t.} \quad & \text{(a): } xs_{\min} \leq x_n \leq xs_{\max}, \quad \text{(b): } ys_{\min} \leq y_n \leq ys_{\max}, \quad \text{(c): } zs_{\min} \leq z_n \leq zs_{\max}, \quad \forall n \in N_p, \\
 & \text{(d): } \sum_{j=1}^{N_p} x_{ij} \leq 1, \quad \forall i \in N_p, \quad \text{(e): } \sum_{j=1}^{N_p} x_{1j} = \sum_{j=1}^{N_p} x_{jg} = 1, \quad \text{(f): } \sum_{j=1}^{N_p} E_{ij} - \sum_{j=1}^{N_p} E_{ji} = 0, \quad \forall i \in [2 \dots N_p], \\
 & \text{(g): } \sum_{i=1}^{N_p} \sum_{j=1}^{N_p} E_{ij}x_{ij} \leq E_u, \quad \text{(h): } \sum_{i=1}^{N_p} \sum_{j=1}^{N_p} \{x_{ji} - x_{ik} \leq y_i\}, \quad \forall k \in [2 \dots N_p], \quad \text{(i): } x_{ij}, y_i \in [0, 1], \quad \forall i, j \in N_p, \quad (1)
 \end{aligned}$$

Constraints (1.a), (1.b), and (1.c) represent the allowed range for the UAV point of view, where $x_{\min}$ and $x_{\max}$, $y_{\min}$ and $y_{\max}$, and $z_{\min}$ and $z_{\max}$ are the minimum and maximum values of x_n , y_n , and z_n , respectively. Constraint (1.d) ensures that at maximum only one link exists from location i to location j . Constraint (1.e) ensures starting and end location constraints. Constraint (1.f) ensures the flying energy from location i to location j is the same as flying energy from location j to location i . Constraint (1.g) ensures the flying energy from location i to location j should be sufficient. Constraint (1.h) ensures the intermediate location constraint which indicates whether a location i is in waypoints of the path. Constraint (1.i) ensures that x and y are binary variables.

2.3. Smoothing strategy

The path generated by the proposed algorithm consists of waypoints. However, the path segments connecting these waypoints are composed of linear segments, making it challenging for a specific UAV to track them smoothly. Furthermore, taking turns consumes a significant amount of energy for a drone. Therefore, it becomes necessary to enhance the path by making it smoother, ensuring suitability for the UAV. The primary objective is to generate a path that adheres to the maximum curvature constraint of the UAV. To address this issue, we apply a quadratic Bezier curves to the generated segments, smoothing them out and reducing the impact of turns on energy consumption. Fig. 3 shows the control points and provides a schematic representation of the path-smoothing strategy.

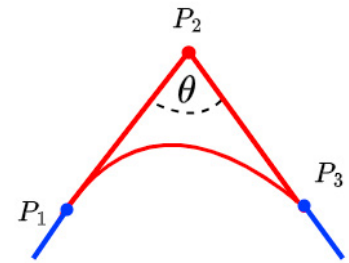


Fig. 3: Path smoothing strategy.

3. Results and Flight Simulations

In order to assess the effectiveness of our proposed approach in efficiently avoiding obstacles while minimizing energy consumption, we conducted a simulation using the Matlab tool. The simulation includes various scenarios involving different obstacles to demonstrate the effectiveness of the proposed algorithm. Additionally, we examined the influence of the obstacles on energy consumption by varying the number of obstacles from 0 to 8 obstacles. The simulation is executed on PC, Intel Core i7 CPU with 16 GB memory. The specific parameters used for the simulation can be found in Table 1.

Table 1: Parameters setting

Area	Static Obs	Dynamic Obs	UAV altitude	Segment size	Regulation constraints (z_{min}, z_{max})
100 m x 100 m	9, 15	15	50 m	20 m	[30 m to 120 m]

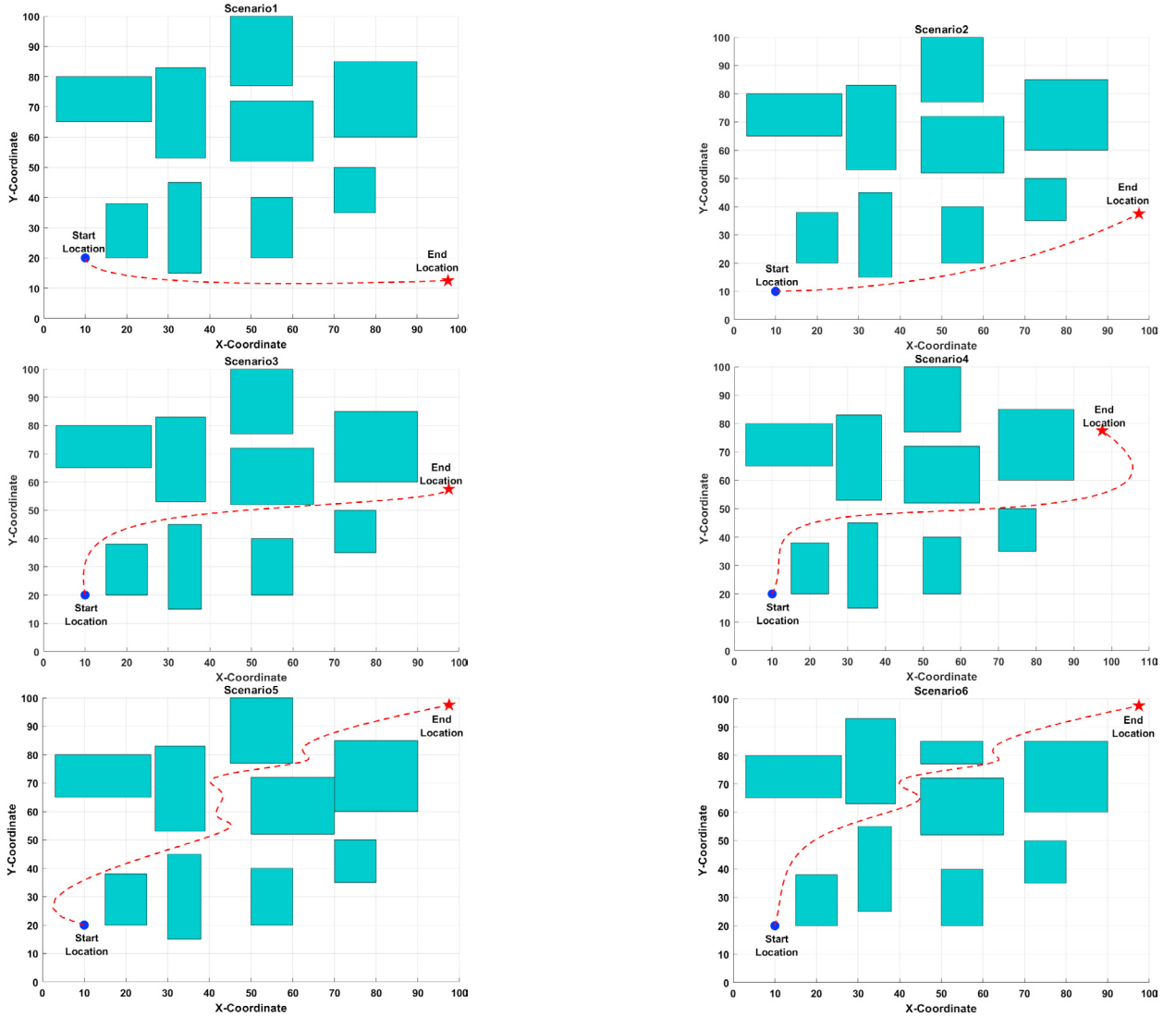


Fig. 4: Two dimensional view of UAV paths for different scenarios.

3.1. UAV path planning performance

The drone flies from starting position to the destination and generates its path with minimize energy consumption while ensuring avoidance of obstacles. In the scenarios, the obstacles are randomly generated across the environment, creating a narrow passages for UAV to navigate. Scenarios in Fig. 4 illustrate different 2D views of UAV paths. Figures show narrow passage, especially in scenarios 3, 4, 5, and 6. The aim is to test the ability of the proposed approach to navigate through such narrow passages. It can be easily observed that the proposed approach successfully generates collision-free paths in all scenarios.

3.2. Energy consumption performance

Energy consumption is a crucial metric for evaluating the performance of the algorithm. It demonstrates the ability of algorithm to generate a collision-free path while minimizing energy consumption. Without loss of generality, we normalized the energy consumption by dividing the total energy consumed over the optimal energy required when a drone follows the shortest path from the starting location to the destination at its maximum speed. Different environmental scenarios are simulated, and the normalized average energy consumption during the mission is illustrated in Fig. 5. Notably, different scenarios depict different energy consumption due to different obstacles along the path from starting point to the destination.

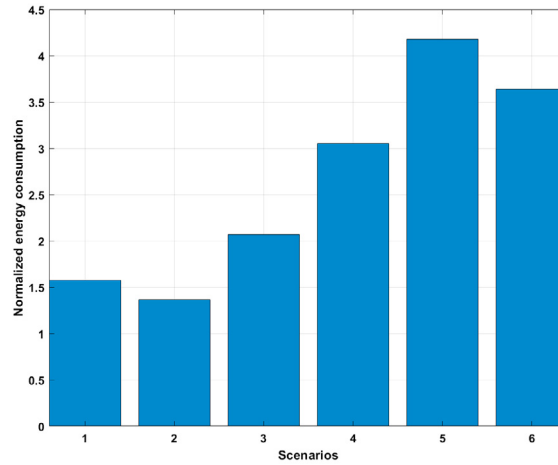


Fig. 5: Two dimensional view of UAV paths for different scenarios.

To further optimize energy consumption, we conduct a study on the impact of obstacles on energy consumption. In the scenario, we apply a smoothing strategy to the generated path and compare the results with paths that are not smoothed. We vary the number of obstacles from 0 to 8, and record the normalized energy consumption with and without the smoothing approach. The results, depicted in Fig. 6 (a), clearly show that the smoothing path exhibits lower energy consumption and improves the performance of our proposed approach.

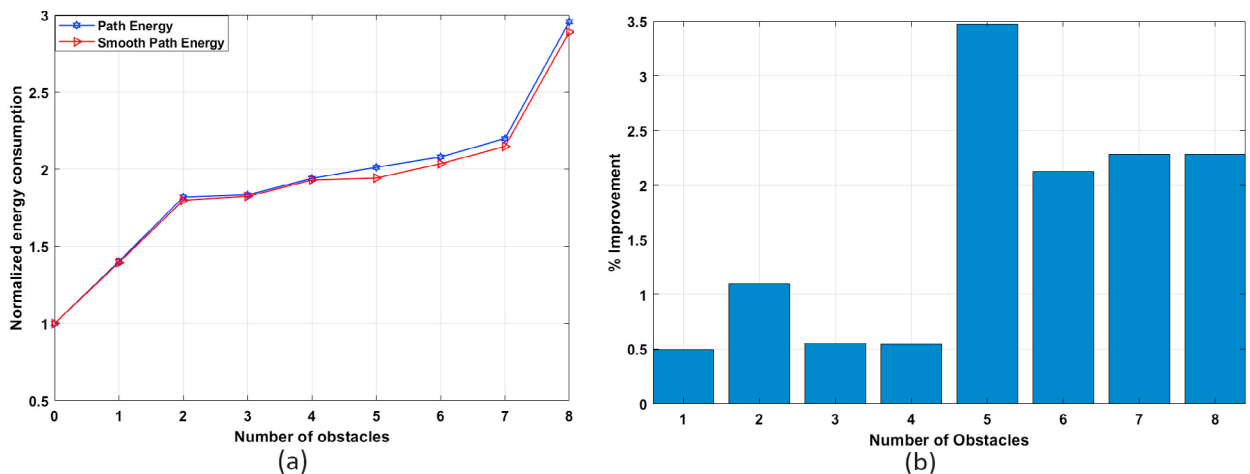


Fig. 6: (a) Normalized energy consumption with respect to the number of obstacles. (b) Percentage improvement in energy consumption when employing the smoothing approach.

To further demonstrate the impact of the smoothing strategy on energy consumption, we plot the percentage improvement in Fig. 6 (b). It is evident that there is a notable improvement in energy consumption. The reason behind this improvement is that sharp turns require the drone to decelerate and accelerate, which consumes additional energy. The smoothing approach mitigates the effect of these sharp turns, leading to reduced energy consumption.

4. Conclusion

In this study, we proposed an online energy-efficient path planning approach for UAV in complex environments using Mixed Integer Linear Programming (MILP) and a path smoothing strategy. The path planning problem was formulated as an MILP, incorporating a cost function to minimize energy consumption while ensuring terrain obstacle avoidance. By utilizing Receding Horizon Control (RHC) and optimization strategies, the path was divided into segments with constraints to avoid collisions with obstacles. This approach enabled fast and accurate navigation in dense environments, ensuring a collision-free path. The results demonstrated the effectiveness and accuracy of the proposed approach in complex environments with a high risk of collision. The implementation showed significant improvements in energy efficiency across all tested scenarios, proving the capability of the method to handle dense obstacle fields effectively.

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References

- [1] T. Sheltami, G. Ahmed, A. Yasar, An optimization approach of iod deployment for optimal coverage based on radio frequency model, CMES-COMPUTER MODELING IN ENGINEERING & SCIENCES (2024).
- [2] Y. Zuo, R. Tharmarasa, R. Jassemi-Zargani, N. Kashyap, J. Thiyaalingam, T. T. Kirubarajan, Milp formulation for aircraft path planning in persistent surveillance, IEEE Transactions on Aerospace and Electronic Systems 56 (5) (2020) 3796–3811.
- [3] G. Ahmed, T. Sheltami, A. Mahmoud, A. Yasar, Iod swarms collision avoidance via improved particle swarm optimization, Transportation Research Part A: Policy and Practice 142 (2020) 260–278.
- [4] A. Majeed, S. O. Hwang, A multi-objective coverage path planning algorithm for uavs to cover spatially distributed regions in urban environments, Aerospace 8 (11) (2021) 343.
- [5] I. H. Imran, N. M. Alyazidi, A. Eltayeb, G. Ahmed, Robust adaptive fault-tolerant control of quadrotor unmanned aerial vehicles, Mathematics 12 (11) (2024) 1767.
- [6] G. A. Ahmed, T. R. Sheltami, A. S. Mahmoud, M. Imran, M. Shoaib, A novel collaborative iod-assisted vanet approach for coverage area maximization, IEEE Access 9 (2021) 61211–61223.
- [7] G. Ahmed, T. Sheltami, A safety system for maximizing operated uavs capacity under regulation constraints, IEEE Access (2023).
- [8] B. Alidaee, H. Wang, F. Landram, A note on integer programming formulations of the real-time optimal scheduling and flight path selection of uavs, IEEE transactions on control systems technology 17 (4) (2009) 839–843.
- [9] N. Nigam, S. Bieniawski, I. Kroo, J. Vian, Control of multiple uavs for persistent surveillance: Algorithm and flight test results, IEEE Transactions on Control Systems Technology 20 (5) (2011) 1236–1251.
- [10] Y. Shen, G. Fan, Rhc method based 2d-equal-step path generation for uav swarm online cooperative path planning in dynamic mission environment, in: Proceedings of the 2023 3rd International Conference on Artificial Intelligence, Automation and Algorithms, 2023, pp. 1–9.
- [11] J. S. Bellingham, M. Tillerson, M. Alighanbari, J. P. How, Cooperative path planning for multiple uavs in dynamic and uncertain environments, in: Proceedings of the 41st IEEE Conference on Decision and Control, 2002., Vol. 3, IEEE, 2002, pp. 2816–2822.
- [12] S. Ragi, H. D. Mittelmann, Mixed-integer nonlinear programming formulation of a uav path optimization problem, in: 2017 American Control Conference (ACC), IEEE, 2017, pp. 406–411.
- [13] G. Ahmed, T. Sheltami, A. Mahmoud, A. Yasar, Energy-efficient uavs coverage path planning approach, CMES-COMPUTER MODELING IN ENGINEERING & SCIENCES 136 (3) (2023) 3239–3263.
- [14] C. Di Franco, G. Buttazzo, Coverage path planning for uavs photogrammetry with energy and resolution constraints, Journal of Intelligent & Robotic Systems 83 (3) (2016) 445–462.
- [15] G. Ahmed, T. Sheltami, M. Ghaleb, M. Hamdan, A. Mahmoud, A. Yasar, Energy-efficient internet of drones path-planning study using meta-heuristic algorithms, Applied Sciences 14 (6) (2024) 2418.
- [16] Z. AlMania, T. Sheltami, G. Ahmed, A. Mahmoud, A. Barnawi, Energy-efficient online path planning for internet of drones using reinforcement learning, Journal of Sensor and Actuator Networks 13 (5) (2024) 50.